



Predicting the Influence of Music Education on Academic Performance in Children

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Introduction

Background - Music education is often associated with academic performance and cognitive development. Prior research is mixed: some studies find academic or cognitive benefits, while others suggest that observed relationships may reflect household resources, socioeconomic status, and selection effects.

Data - This project uses data from the Panel Study of Income Dynamics - Child Development Supplement for 2014, 2019, and 2021. The final sample included 1,173 students with valid normalized GPA results.

Question - To what extent is music education associated with academic performance, and do music-related variables meaningfully predict student GPA?

Hypothesis - Music-related access and participation are expected to have a positive association with GPA, but the relationship is expected to be small because academic performance is shaped by many broader factors.

Methods

The analysis compared music-related variables across three survey years using both statistical testing and predictive modeling.

Statistical tests used:

- Descriptive statistics for GPA distributions
- Independent samples t-tests for GPA differences between participation groups
- Point-biserial correlations between binary music variables and GPA
- Ordinary Least Squares regression for the combined dataset
- Train/test linear regression for predictive modeling

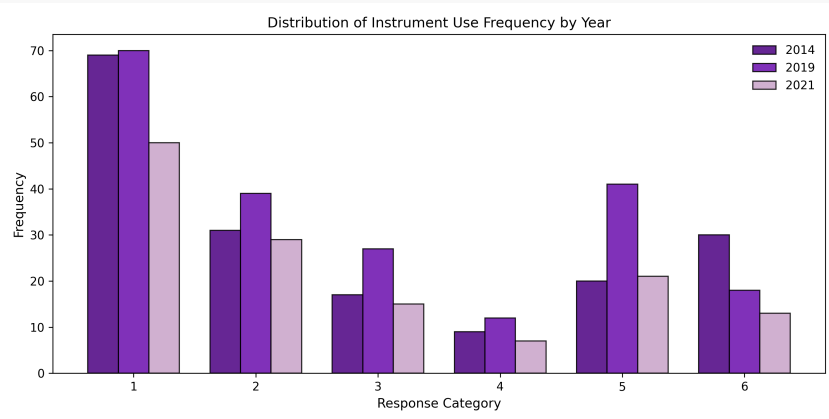
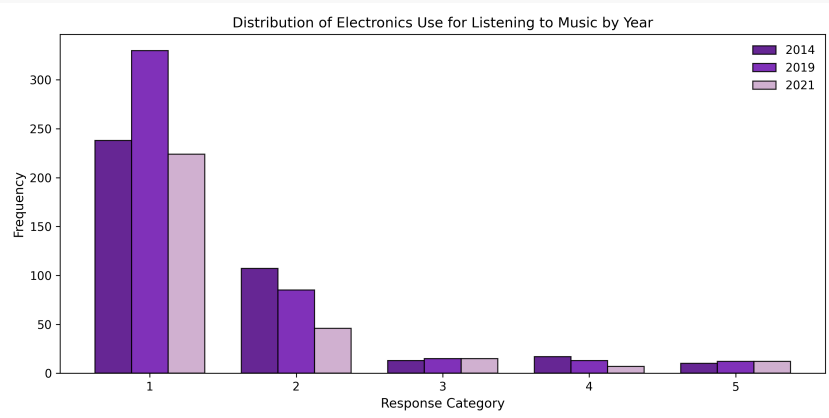
Predictors of academic performance:

- Music lessons and general music participation
- Instrument access and instrument-use frequency
- Arts tutoring
- Community and religious art groups participation

Music Engagement Variables

The ordinal variables describe the intensity of music engagement.

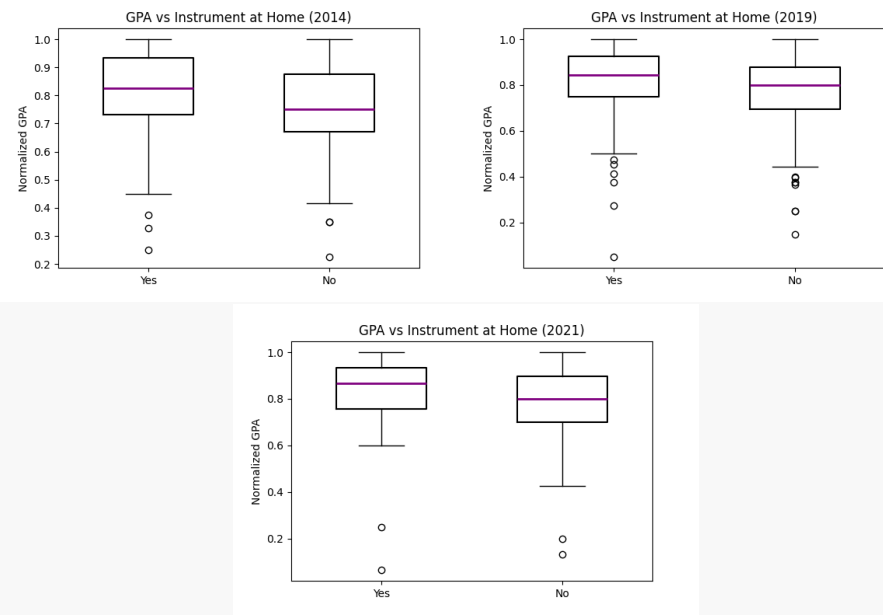
- The use of electronics for listening to music was highly concentrated in the most frequent response category across all three years, suggesting that this behavior was nearly universal among students.
- Instrument-use frequency showed greater variation across response categories, making it a more useful measure of active music engagement.



Key Variable - Instrument at Home

The most consistent results across the analysis came from the relationship between GPA and the presence of a musical instrument in the home.

- In 2014, 2019, and 2021, students with an instrument at home tended to have higher median GPA values than students without one.
- This pattern appeared across the t-tests, correlation analysis, and combined regression model, making instrument availability the most stable variable examined.

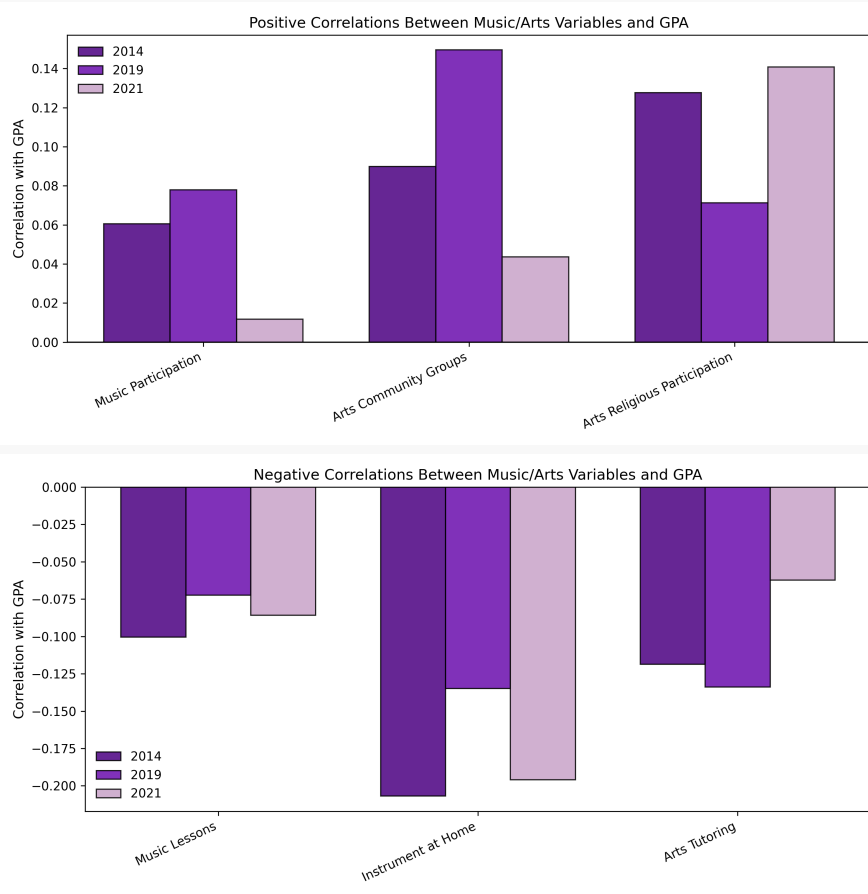


Correlations with GPA

Point-biserial correlations were small in magnitude, indicating weak relationships between individual music or arts variables and GPA.

- Instrument availability showed the most stable association across all three years.
- Arts tutoring showed negative associations in 2014 and 2019.
- Community and religious arts participation showed small positive associations in selected years.

These results suggest that arts engagement may matter, but the strength and direction of the relationship varies across cohorts.



Regression Results

A pooled Ordinary Least Squares (OLS) regression tested whether music-related variables remained associated with GPA when considered simultaneously. Performance was weak across all variables, with low R^2 value of 0.059. The model explained only 5.9% of GPA variation, indicating that the selected variables capture only a small part of academic performance.

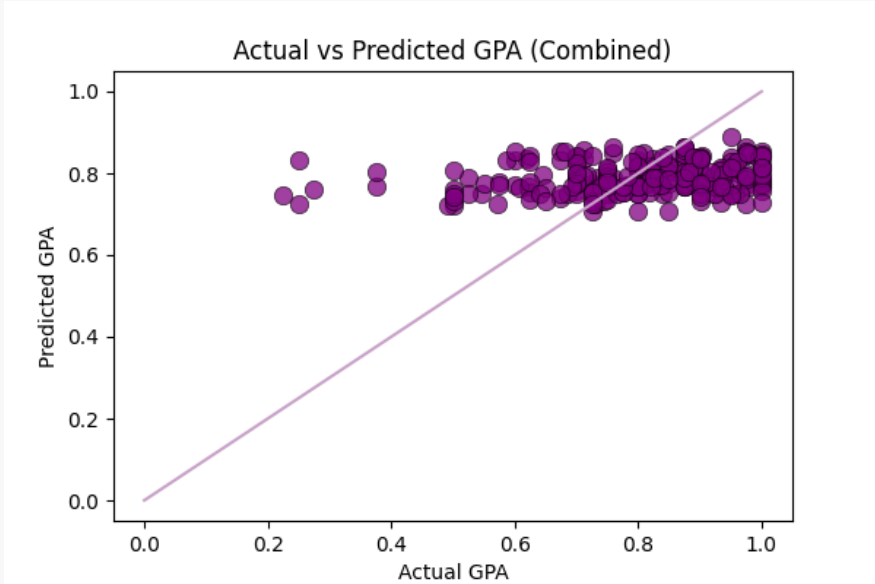
Statistically significant predictors:

- Instrument at home
- Arts tutoring
- Religious and community groups arts participation
- Survey year

Direct music participation variables, including music lessons and general music participation, were not statistically significant after accounting for the full set of predictors.

Predictive Model

Predictive modeling tested whether music-related variables could estimate the GPA of students. After dividing the dataset into a training and testing split, the GPAs of the students not included in the training sample were predicted. Performance was weak across all datasets, with low R^2 values and predictions clustering near the average GPA, showing limited ability to separate lower- and higher-performing students.



Conclusion

Music-related variables are associated with GPA, but the relationships are small. The strongest and most consistent result is not direct music participation, but access to a musical instrument in the home. This suggests that household context and access to resources may play an important role in the relationship between music education and academic performance.

Music education appears to matter as part of a broader system of support, opportunity, and socioeconomic status, rather than as a strong standalone predictor of GPA.

Limitations and Future Work

Limitations

- Observational data cannot establish causality.
- Binary variables do not capture duration, intensity, or quality of music participation.

Future Work

- Include stronger socioeconomic controls, such as household income and parental education.
- Use richer music engagement measures, including years of study, practice frequency, and type of instruction.